The (De-)evolution of Evolution Games

A Content Analysis of the Representation of Evolution Through Natural Selection in Digital Games

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1 Abstract

MC 451 · Research Methods in Mass Media · Spring 2026 · INSTRUCTOR EXAMPLE

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About This Example

This portfolio is an **instructor-built example** of the MC 451 final project, adapted from a published study: Leith, Ratan, and Wohn (2016), *The (De-)evolution of Evolution Games: A Content Analysis of the Representation of Evolution Through Natural Selection in Digital Games* in the *Journal of Science Education and Technology* (Leith et al., 2016). The dataset (N = 22) is the real coding data reported in Table 2 of that paper. The structure has been adapted to fit the MC 451 final-portfolio template, which uses Chi-Square as the inferential test in place of the original paper's qualitative-thematic analysis. Use this example to see what the final deliverable looks like end-to-end.

Abstract

This study examines how digital games portray the theory of evolution through natural selection (ENS), the cornerstone of the biological sciences. Using quantitative content analysis of 22 evolution-themed digital games (14 commercially distributed, 8 academically developed for education or research), this study addressed two research questions: (1) what proportion of evolution-themed games contain none, some, or all three principles of ENS — phenotypic variation, differential fitness, and fitness heritability — and (2) does the completeness of ENS representation differ between commercial and academic games? Findings indicate that only 5 of 22 games (22.7%) presented all three principles, with phenotypic variation by far the most commonly depicted single principle. A Chi-Square test of independence comparing ENS completeness against game origin produced a non-significant result; however, several expected cell counts fell below the conventional threshold of 5, limiting the inferential strength of the test. These findings support the broader argument that the interactive properties that make digital games engaging — particularly direct player control over outcomes — are inherently in tension with the automatic, non-agentive process of natural selection.

1.0.1 About This Study

This portfolio presents a content analysis of digital games that represent biological evolution. The study traces which scientific principles of evolution through natural selection (phenotypic variation, differential fitness, fitness heritability) appear in commercially distributed video games versus games developed by educators and researchers. The motivating question: when a teacher reaches for a "fun" game to introduce evolution, are those games actually teaching the science correctly?

1.0.2 How to Navigate This Report

Section	What You Will Find
Introduction	Background on ENS, the Mapping Principle, and the formal research questions
Methodology	Database selection, keyword search, sampling decisions, and the three-principle codebook
Results	Frequency of the three ENS principles, distribution of completeness levels, and the Chi-Square test
Discussion	Interpretation, the diminishing-returns finding, limitations, and the small-sample caveat
References	Full bibliography

2 Introduction

Phase I · The Journalist

Science, technology, engineering, and mathematics (STEM) education increasingly relies on informal learning supplements — videos, simulations, and digital games — to reach students with concepts that are too abstract, too slow-moving, or too dangerous to demonstrate directly in a classroom. Among these supplements, digital games have grown in popularity as inquiry-based educational tools (Lieberman, 2012; Ritterfeld et al., 2009). Evolution through natural selection (ENS) is one such concept: foundational to the biological sciences (Dobzhansky, 1973), persistently misunderstood by the public, and apparently well suited to the slow, accumulating, generations-long timescale that a digital simulation can compress into a few minutes of play. Yet whether games that *describe* themselves as evolution-related actually *teach* the science of evolution is an open empirical question — and one with real consequences for any teacher who reaches for a game in good faith.

2.1 Background and Significance

Prior research on educational digital games has established that games can convey complex scientific content through inquiry-based learning (Kennedy-Clark & Thompson, 2011; Reimann & Thompson, 2009). Inquiry-based learning, as articulated by Schwab (1960) and elaborated by Rezba et al. (1999), ranges from confirmation inquiry (verifying a known result) through structured, guided, and open inquiry (formulating and testing original questions). Digital games are uniquely positioned to support these modes because they can present dynamic systems whose internal logic the player must discover through play.

However, the educational value of any specific game depends on whether the game's mechanics actually map onto the real-world phenomenon it claims to teach. This conditional has been formalized as the **Mapping Principle** (Williams, 2010): digital experiences translate to offline behavior, knowledge, and reasoning to the extent that the digital representation closely parallels its physical counterpart. A game like *Angry Birds* may give players a rough sense of projectile motion but does not formally teach physics. The question for evolution games is analogous: which games' mechanics map onto the actual mechanism of natural selection, and which only borrow the vocabulary?

Despite the steady production of "evolution-themed" titles for the consumer market, the current state of evolution games is largely uncatalogued. Some games may even contribute to misconceptions (Bean et al., 2010), teaching players that evolution is a deliberate,

agentive process driven by player choice rather than the automatic consequence of differential reproductive success. This study addresses that gap by systematically coding the population of available evolution-themed games against the established scientific definition of natural selection.

2.2 Theoretical Framework

This study is grounded in the three-principle definition of evolution through natural selection articulated by Lewontin (1970), building on Darwin (1859). ENS occurs when three conditions are simultaneously met:

1. **Phenotypic variation** — organisms within a population possess different traits.
2. **Differential fitness** — those trait variations produce different rates of survival and reproduction.
3. **Fitness heritability** — the advantageous traits of survivors are passed to offspring.

When all three conditions hold, natural selection follows automatically; the system requires no external designer or director. Conversely, if any of the three is missing — for instance, if all variations have equal fitness, or if traits do not transmit to offspring — the system does not produce evolutionary change.

The Mapping Principle (Williams, 2010) supplies the second theoretical thread. The principle predicts that a digital game can teach ENS only if its mechanics simulate all three Lewontin conditions. A game in which players manually decide which organisms reproduce, or in which all organisms survive equally regardless of trait, does not map onto natural selection — it maps onto artificial selection or onto something else entirely. Together, Lewontin's three principles supply the *what* of accurate ENS representation, and the Mapping Principle supplies the *why* of why mapping fidelity matters for learning.

2.3 Research Questions

Based on the foregoing, this study addresses the following research questions:

> **RQ1:** What proportion of evolution-themed digital games contain none, some, or all three principles of evolution through natural selection?
>
> **RQ2:** Does the completeness of ENS representation differ between commercially distributed and academically developed games?

Together, these questions produce a replicable, principle-by-principle account of which evolution-themed games meet the scientific criteria of natural selection and which do not — moving beyond intuitive impressions toward a defensible audit of the available teaching materials.

3 Methodology

Phase II–III · The Architect & Builder

This study employed systematic quantitative content analysis to address the two research questions introduced in the previous section. Content analysis is appropriate here because it provides a replicable, transparent method for identifying patterns in media content at a scale that is not feasible through close reading alone.

3.1 Data Source

Because no single comprehensive database catalogues all digital games (in the way IMDB.com catalogues films), data were drawn from the six largest publicly accessible video-game databases as ranked by monthly unique visitors and total searchable game count: IGN.com, GameFAQs.com, GameSpot.com, GiantBomb.com, GamerDNA.com, and MobyGames.com. Database rankings were verified using the Compete (2012) web traffic measurement service. These six sources were used in combination because no single source returned an exhaustive list of evolution-themed titles, and each contributed distinct entries to the final sample.

A second, supplemental data collection phase was conducted to capture games that the commercial databases routinely exclude — specifically academically developed games made for teaching or research. This second phase used Google web searches and outreach to other researchers and educators who work with so-called “serious games.”

3.2 Sample and Sampling Procedure

The keyword **“evolution”** was used to search each of the six databases. Related search terms (e.g., *evolve*, *natural selection*) were tested but produced no additional unique data, irrelevant data, or redundant data. Initial keyword counts varied widely across databases, from 52 results (GiantBomb) to 1,500 results (GameSpot, prior to refinement). GameSpot’s initial result set was uniquely inflated by the use of the term *evolution* in unrelated game-sequel titles (e.g., *Trials Evolution*, *Pac-Man Evolution*, sports-game iterations). A secondary search using exclusionary terms (Trials, Pac-Man, soccer, sports, skateboarding, hackers, revolution, Tron) reduced GameSpot’s results to 269 games.

From the six resultant datasets, every game was examined for substantive relationship to evolution as a biological process. Games whose only connection to "evolution" was a sequel-naming convention or a brand pun were excluded. This screening produced a final Study 1 sample of **14 games** spanning the years 1982 to 2012 across ten platforms.

The supplemental phase produced an additional **8 games** identified through Google searches (`evolution + game`, `evolve + game`, `natural selection + game`) and personal-network inquiries to colleagues working in serious-games research and STEM education.

The combined final sample consisted of **N = 22 games**, each constituting one unit of analysis: a single distinct evolution-themed digital game title.

3.3 Variables and Measures

Game Origin is the supplemental administrative variable in this analysis. It distinguishes between **Commercial** games (those discoverable through the major game databases used in Study 1, n = 14) and **Academic** games (those identified only through Google searches or personal networks in Study 2, n = 8). Origin captures the practical difference between games that a teacher or learner might find through ordinary commercial channels and those that require connections to research networks to discover.

Phenotypic Variation refers to the appearance of distinct organisms with varying digital physiologies or traits. Per Lewontin's (1970) first principle of natural selection, both intraspecies variation (different individuals within the same species) and interspecies variation (different species) qualify. A game was coded **Yes** when characters within the game world exhibited observable trait differences (e.g., body shape, coloration, behavior) that the game's mechanics treated as meaningful, and **No** when the game presented uniform organisms with no observable trait differences.

Differential Fitness refers to the capacity for character or species survival to be contingent on traits, per Lewontin's second principle. A game was coded **Yes** when otherwise-similar organisms experienced different rates of survival or success that depended on their traits, and **No** when player intervention (manual control of which organisms survive or breed) overrode trait-based outcomes. The most common reason for a "No" coding was the player's ability to directly manipulate any specific organism's success regardless of its fitness.

Fitness Heritability refers to the presence of organisms passing traits to offspring within the game, per Lewontin's third principle. A game was coded **Yes** when reproduction occurred within the game's mechanics *and* offspring exhibited observable similarity to their parents (visual, behavioral, or both), and **No** when reproduction was either absent or treated as a reset (offspring with no inherited resemblance to parents).

ENS Completeness is a derived variable computed from the three principle-presence codings. Games were classified as **None** (zero principles present), **Partial** (one or two

principles present), or **Complete** (all three principles present). This three-level variable serves as the primary categorical outcome (`var1`) for the analyses in the next chapter.

3.4 Coding Procedure and Codebook Refinement

Coding was conducted by committee. One author collected and reviewed all available material for each candidate game — including in-game playthroughs where possible, and developer descriptions, gameplay videos, screenshots, and reviews where direct play was not possible (some titles required outdated or region-locked hardware). The full author team then coded the games together against the three principles until unanimous classifications were reached. This procedure provided a single, jointly informed judgment for each game; the qualitative and exploratory nature of the work, combined with the small sample size, did not require a separate inter-coder reliability assessment with external validators. The committee approach was used in preference to independent coding because the principal coding ambiguities involved interpretation of game mechanics across very different game eras and platforms — an interpretation problem better resolved by deliberation than by reliability statistics.

4 Results

Phase IV · The Analyst

This section presents the findings of the content analysis (N = 22 games). Results are organized to address each research question in turn. Descriptive statistics are presented first, followed by inferential analysis.

4.1 Descriptive Statistics

4.1.1 Overview of the Sample

Game Origin	N	% of Sample
Commercial	14	63.6%
Academic	8	36.4%

Table 4.1: Table 1. Sample composition by game origin (commercial vs. academic).

The sample comprises 22 evolution-themed digital games. Just over six in ten (63.6%) were commercially distributed games discoverable through standard video-game databases; the remaining 36.4% were academically developed games identified only through targeted web searches and personal-network outreach. This imbalance is itself a finding: the commercial-channel discoverability of academic evolution games is substantially weaker than that of mainstream commercial titles.

4.1.2 Distribution of Primary Variable: ENS Completeness

ENS Completeness	N	Percent	Cumulative %
None	2	9.1%	9.1%
Partial	15	68.2%	77.3%
Complete	5	22.7%	100%

Table 4.2: Table 2. Frequency distribution of ENS Completeness across all coded games (N = 22).

Table 2 addresses RQ1 directly. The dominant category is **Partial** representation (15 of 22 games, 68.2%) — games that present one or two of Lewontin's three ENS principles but not all three. Only 5 games (22.7%) presented the full set of principles required for ENS to actually occur. Two games (9.1%) — both academically developed information-presentation titles — presented none of the three principles in their game mechanics, despite their explicit framing as evolution education.

4.1.3 Bar Chart: Distribution of ENS Completeness

Figure 1 visualizes Table 2. Two patterns stand out. First, Partial representation is the modal category by a wide margin — most evolution-themed games depict some but not all of the principles required for natural selection to function as a closed system. Second, the Complete category is sparse: only 5 of 22 games (22.7%) clear the bar of representing all three Lewontin principles in their mechanics.

4.1.4 Distribution by Individual Principle

Principle	Games (Yes)	% of N=22
Phenotypic Variation	20	90.9%
Differential Fitness	6	27.3%
Fitness Heritability	9	40.9%

Table 4.3: Table 3. Presence of each Lewontin principle of natural selection across the 22 coded games.

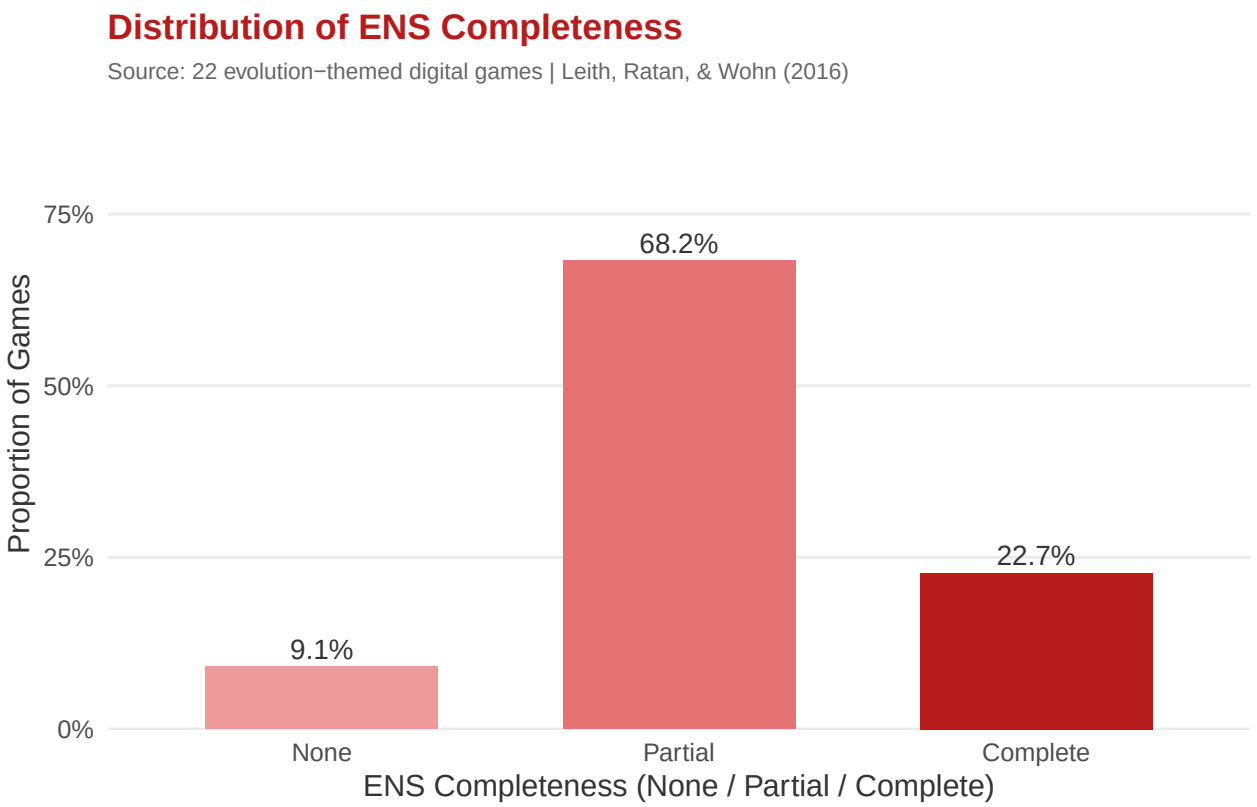


Figure 4.1: Figure 1. Distribution of ENS Completeness across all coded games (N = 22). The Partial category dominates, while only one game in five contains all three Lewontin principles required for natural selection to occur.

Table 3 disaggregates ENS Completeness into the three principles. **Phenotypic variation** appears in 20 of 22 games (90.9%) — by far the most commonly depicted principle. **Differential fitness** and **fitness heritability** appear in only 6 (27.3%) and 8 (36.4%) games respectively. This pattern suggests that game designers reach naturally for visible diversity (different-looking organisms) but more rarely simulate the consequence-and-inheritance machinery that turns variation into actual evolution.

4.2 Inferential Analysis

4.2.1 Chi-Square Test of Independence: ENS Completeness × Game Origin

This Chi-Square test addresses RQ2: does the completeness of ENS representation differ between commercially distributed and academically developed games?

	Game Origin	
ENS Completeness	**Commercial**	**Academic**
None	0	2
Partial	11	4
Complete	3	2

Table 4.4: Table 4. Cross-tabulation of ENS Completeness by Game Origin (N = 22).

Table 4 displays the cross-tabulation. The two **None** games are both academic; commercial games have at least one principle in every case. The **Complete** category is split — 3 commercial games (*Creatures*, *SimEarth*, *SimLife*) and 2 academic games (*Selection Game*, *Who Wants to Live a Million Years*) — and the **Partial** category is dominated by commercial games (11 of 15).

Chi-Square Result:

A Chi-Square test of independence was conducted to examine the relationship between ENS Completeness and Game Origin. The result was not statistically significant, $\chi^2(2) = 4.14$, $p = 0.126$.

Three observations on the Chi-Square result:

1. **The test was not statistically significant at α = .05** ($p = 0.126$). The data do not provide evidence of a relationship between game origin and ENS completeness. Both commercial and academic games appear roughly comparable in the proportion of titles that present all three Lewontin principles.

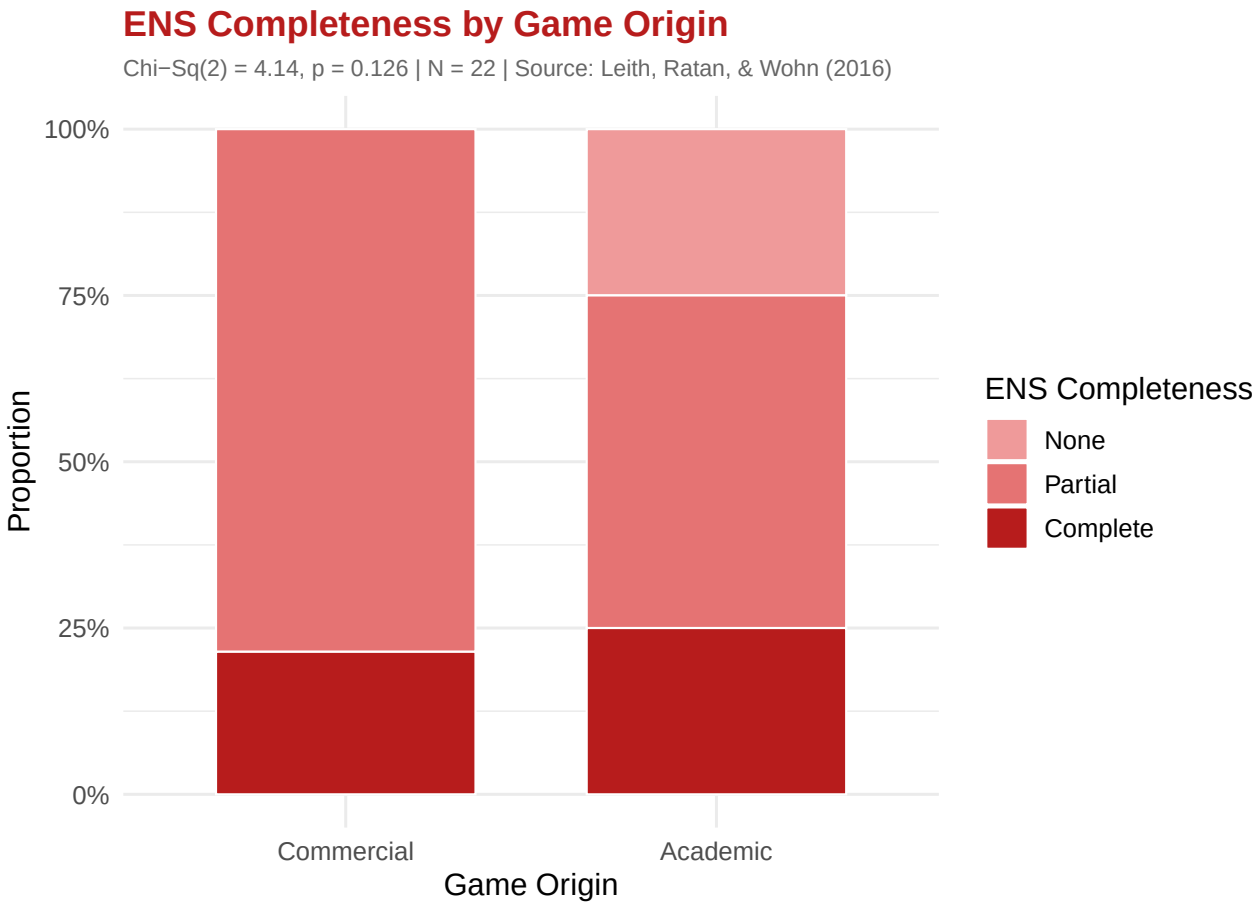


Figure 4.2: Figure 2. Stacked-bar plot showing the relationship between Game Origin and ENS Completeness. Bars represent the proportional composition of each origin group; cell counts are small, particularly for the None and Complete categories, which limits the inferential strength of the Chi-Square test reported above.

2. **All academic games rated as "None" (n = 2) belong to a specific subtype** — quiz-based information-presentation titles (*Angry Aliens: Evolution* and *Life Preservers*) that teach evolution as content rather than simulating it as a process. The qualitative significance of this pattern is not captured by the inferential test alone.
3. **The minimum expected cell count was 0.73**, which falls below the conventional threshold of 5 required for the Chi-Square test of independence. Three of the six expected cell counts in this 3×2 table are below 5. This is a consequence of the small sample size (N = 22) and the rarity of the Complete and None categories. **The Chi-Square result should therefore be interpreted with substantial caution** — Fisher's Exact Test or a substantially larger sample would be required to test this relationship rigorously. The original published study used qualitative thematic analysis precisely because the available population of evolution-themed games was too small to support inferential statistics.

4.3 Summary of Findings

i Key Findings at a Glance

- **RQ1** (How complete is ENS representation in evolution-themed games?): The dominant category is Partial (15 of 22 games, 68.2%). Only 5 games (22.7%) present all three Lewontin principles. Two games (both academic, both information-presentation quiz titles) present none of the three principles in mechanics.
- **RQ2** (Does completeness differ between commercial and academic games?): The Chi-Square test of independence was not statistically significant. However, expected cell counts below 5 in three of six cells limit the inferential strength of this test substantially. The data do not provide statistical evidence of a relationship; the small sample size precludes confident conclusion either way.

5 Discussion & Conclusion

Phase V · The Publisher

This study examined how digital games portray the theory of evolution through natural selection (ENS). Using quantitative content analysis of N = 22 evolution-themed games — 14 commercially distributed and 8 academically developed — the study addressed two research questions concerning the completeness of ENS representation and whether that completeness differed by game origin. The findings are discussed below in relation to each research question, the broader theoretical framework, and the methodological caveats raised in the previous chapter.

5.1 Addressing the Research Questions

RQ1 asked what proportion of evolution-themed digital games contain none, some, or all three principles of ENS. The frequency distribution presented in Table 2 and Figure 1 revealed that the dominant category is Partial (15 of 22 games, 68.2%), with only 5 games (22.7%) presenting all three Lewontin principles and 2 games presenting none. Table 3 disaggregates the result by individual principle: phenotypic variation appears in 90.9% of games, while differential fitness and fitness heritability appear in only 27.3% and 36.4% respectively.

This pattern has a clear interpretation in light of the Mapping Principle (Williams, 2010): the most visually obvious principle of ENS — different organisms with different traits — is also the easiest to depict in a digital game, and games depict it readily. The two principles that require the *system* of natural selection to actually function as a closed loop — fitness consequences and inheritance — are far rarer. In effect, most evolution-themed games show what evolution *looks like* without simulating how it *works*. The Mapping Principle predicts that this asymmetry will produce learners who can recognize evolutionary outcomes but not reason about evolutionary mechanism.

RQ2 asked whether ENS completeness differed between commercial and academic games. The Chi-Square test of independence was not statistically significant. As Table 4 shows, both commercial and academic games produced Complete-rated titles at roughly comparable rates (3 of 14 commercial, 2 of 8 academic). The two None-rated games are both academic, but they share a specific subtype: information-presentation quiz games (*Angry Aliens: Evolution*, *Life Preservers*) that teach evolution as content rather than simulating it as a process.

The most important caveat for RQ2 is statistical: with N = 22 and three of six expected cell counts below the conventional Chi-Square threshold of 5, the inferential test is fragile. The null result reported here should not be read as evidence that commercial and academic games are equivalent — only that the available data are insufficient to support a confident conclusion either way. The original published study used qualitative thematic analysis precisely because the population of evolution-themed games is small enough that descriptive and qualitative methods are more honest than inferential ones.

5.2 Theoretical Implications

Taken together, these findings complicate a common assumption about the educational potential of "fun" games. The implicit logic of much edu-tainment design holds that if a game *invokes* a topic — sets it as backdrop, names it, lets the player play through it — the game has done its educational work. The present results push back: invoking evolution is widespread in commercial games, but actually *simulating* the mechanism of natural selection is rare, and rare in a particular way. Player agency, the very feature that makes games engaging, is in tension with natural selection's defining property of operating *without* directorial intervention. A game in which the player saves their favorite organism from a fitness consequence is not, mechanically, a game about natural selection.

This finding generalizes beyond evolution. Any educational target whose underlying mechanism requires automaticity — natural selection, market clearing, mass action in chemistry, equilibrium-reaching processes broadly — is in similar tension with the directorial premise of most digital games. The Mapping Principle does not say the tension is fatal; sandbox-style and interactive-storytelling designs (*SimEarth*, *SimLife*, *Selection Game*) exemplify mechanics that allow ENS to play out without player override. But these designs are uncommon, and the dominant commercial game-design idiom of player-as-protagonist works against them.

5.3 Limitations

Several limitations of the present study should be noted. First, the sample size (N = 22) is small. With three Chi-Square cells below the expected-count threshold of 5, the inferential test reported in RQ2 should not be taken as the final word on whether commercial and academic games differ in ENS completeness. Replication with a substantially larger sample, or use of Fisher's Exact Test as the primary inferential tool, would strengthen any conclusion.

Second, no single game database is comprehensive. Flash games, browser games, mobile-only titles, and games for outdated or region-locked hardware are systematically under-represented across the six databases sampled. The Study 2 supplemental phase partly compensated for this — Google searches and personal-network outreach surfaced

games the databases missed — but the supplemental phase was not exhaustive either, and additional ENS-relevant games likely exist outside the present sample.

Third, the coding-by-committee approach used in the original study, while appropriate for a small qualitative analysis, does not produce the inter-coder reliability statistic that a fully quantitative study would normally report. A future replication with two or more independent coders, computing Cohen's κ or Krippendorff's α, would address this.

Fourth, the study coded only for the *presence* of each Lewontin principle, not for the *fidelity* of its depiction. A game that depicts fitness heritability via cartoonish parent-child resemblance is coded the same as one that simulates Mendelian genetics. A more granular fidelity scale would reveal additional structure within the Partial category.

These limitations notwithstanding, the study's systematic sampling procedure across six major databases, its principle-by-principle coding scheme grounded in a foundational biology citation (Lewontin, 1970), and its honest engagement with the small-sample inferential constraints together represent meaningful methodological strengths that support the credibility of the descriptive findings.

5.4 Conclusion

The central finding of this study is that evolution-themed digital games most often invoke evolution without simulating natural selection: phenotypic variation is everywhere, but differential fitness and fitness heritability are rare, and the combined presence of all three Lewontin principles is rare still. This matters because educators and parents reach for "evolution games" assuming that play with the topic produces learning about the mechanism — and the mapping research suggests that without all three principles in the mechanics, that assumption fails. Future research could extend this work by directly testing the learning outcomes of mechanically Complete versus Partial evolution games on student understanding of natural selection, controlling for game polish and entertainment value, and by surveying a substantially larger and more recent sample of mobile and browser-based evolution-themed titles.

A Note on This Example

This portfolio is an instructor-built example for MC 451: Research Methods in Mass Media at Southern Illinois University Edwardsville (Spring 2026). The dataset is the real Table 2 from Leith, Ratan, and Wohn (2016); the structure has been adapted to match the MC 451 final-portfolio template, which uses Chi-Square as its inferential test in place of the original paper's qualitative-thematic analysis. The data, codebook, and full analysis files are available in the GitHub repository linked at the top of this page.

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